

# Biopore effects on phosphorus biogeochemistry in subsoils



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## ABSTRACT

Biopores are characterised by high concentrations of plant available nutrients and provide preferential pathways for root growth into the subsoil, thereby potentially enabling plants to access phosphorus (P) resources located in the subsoil. Here, we sampled biopores from a replicated agricultural field trial in Klein-Altendorf, Germany, to analyse their nutrient composition and P speciation as determined by Hedley sequential extraction and X-ray absorption near edge structure (XANES) spectroscopy. In addition, we analysed the oxygen isotopic composition of HCl P ( $\delta^{18}\text{O}_{\text{HCl P}}$ ) as an indicator of long-term effects of biological P turnover. We found that biopore effects were most pronounced in the subsoil, where the concentration of easily extractable (labile) P tended to be greater in biopores than in bulk soil, as evident in both Hedley sequential extraction and XANES spectroscopy. We assume that these findings result from inputs of organic matter from the topsoil as well as an input of Ca-particles into subsoil biopores by earthworm activity. Biologically cycled P was subsequently precipitated as Ca-P as evident by  $\delta^{18}\text{O}_{\text{HCl P}}$  values close to equilibrium in biopores even at great depths. When incubating bulk soil samples with  $^{18}\text{O}$ -labelled water, however, we observed a significant increase of  $\delta^{18}\text{O}_{\text{HCl P}}$  values in the topsoil, but only small if any changes of  $\delta^{18}\text{O}_{\text{HCl P}}$  values in the subsoil. Thus, biopores present hotspots of P cycling in the subsoil, but the effect of biopores on overall P turnover in the bulk subsoil is limited.

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## 1. Introduction

In agricultural soils, phosphorus (P) is a limiting element for plant nutrition despite being present in large amounts: most P is bound in chemical forms not readily available to plants (Cross and Schlesinger, 1995; Kruse et al., 2015). Also, substantial amounts of P are stored in the subsoil (Kautz et al., 2013; Barej et al., 2014) and are therefore not immediately within the reach of roots of most cereal crops. Here, biopores, i.e. pores created by earthworms or by tap-rooted pre-crops, have often been suggested as a means to increase plant access to these subsoil resources (Passioura, 2002; McKenzie et al., 2009; Kautz et al., 2013).

Beneficial effects of biopores have been assigned, on the one hand, to their role in providing preferential paths for root growth, especially in compacted subsoils, due to lower penetration resistance and better aeration (Stirzaker et al., 1996; Valentine et al., 2012; Gaiser et al., 2013). Root growth in biopores might further be associated with nutrient mobilisation and cycling by

microorganisms of the rhizosphere. On the other hand, earthworms transport organic matter from the soil surface into biopores and line the walls with casts, which exhibit high microbial activity (Don et al., 2008; Uksa et al., 2015). As a result of both, biopores may potentially present hotspots for nutrient acquisition due to enhanced biological cycling of nutrients as well as due to increased availability of nutrients for plant uptake. For P, it has been demonstrated that earthworm casts in biopores are characterised by significantly higher contents of plant available P relative to the bulk soil (Kuczak et al., 2006; Barej et al., 2014) although the activity of P-releasing enzymes such as acid phosphatase is not significantly elevated in these biopore linings (Jégou et al., 2001; Don et al., 2008). However, even as biopores in subsoils may be stable for several decades (Beven and Germann, 1982; Hagedorn and Bundt, 2002), the stimulation of microbial activity and related nutrient cycling might cease soon after roots have decayed or after earthworms have abandoned the pore (Don et al., 2008; Kuzyakov and Blagodatskaya, 2015). As a result, beneficial effects of earthworm activity do not necessarily translate into an increased P availability in the bulk soil (Vos et al., 2014). Also, roots growing along biopores may still be able to grow into the surrounding bulk

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soil in order to make use of P resources at greater depths (Perkons et al., 2014; Han et al., 2015). Beyond the scale of an individual biopore, it thus remains unclear whether and to which extent biopores affect subsoil P cycling in general and at time-scales exceeding the lifetime of a single root or earthworm.

Phosphorus availability is mainly controlled by the chemical bonding forms of P in soils. Sequential extraction procedures such as the one described by Hedley et al. (1982) have been introduced to assess P pools of different chemical speciation. Yet, as these procedures are operational, there are limitations in assigning a defined P fraction to a certain degree of bioavailability; moreover, different sample timing, handling and pre-treatment, as well as the accuracy of organic and inorganic P assessment may alter fraction yields (Cross and Schlesinger, 1995; Negassa and Leinweber, 2009). The above-mentioned limitations can partly be overcome by X-ray absorption near edge structure (XANES) spectroscopy, as it enables an element-specific and non-invasive direct elemental speciation in the solid phase. This approach has been applied, e.g., to determine effects of P fertilisation on P speciation in soils (Beauchemin et al., 2003; Eriksson et al., 2015).

Another powerful tool to trace biological processes involved in the P cycle is the analysis of the oxygen (O) isotopic composition of phosphate ( $\delta^{18}\text{O}_\text{P}$ ) (Frossard et al., 2011). Under ambient environmental conditions the P–O bond in phosphate is stable when biological activity is absent, i.e. there will be no exchange of O atoms with water (Tudge, 1960; Kolodny et al., 1983; O'Neil et al., 2003). An exchange of O atoms will only occur in the presence of enzymes. In the soil environment, some enzymes, especially phosphohydrolases, are exuded by plants and microorganisms to promote the mineralisation of organic phosphate ( $\text{P}_\text{o}$ ) (Zimmermann et al., 2003; Richardson et al., 2011). During this process, O atoms from the surrounding soil water will be incorporated into the newly formed phosphate molecule. This has been shown for phosphomonoesterases (Liang and Blake, 2006; von Sperber et al., 2014), phosphodiesterases (Liang and Blake, 2009) and phytases (von Sperber et al., 2015; Wu et al., 2015). Furthermore, the activity of the ubiquitous intracellular enzyme pyrophosphatase leads to a complete O exchange between phosphate and water (Chang and Blake, 2015; von Sperber et al., 2016), and thus to a complete equilibration of O isotope ratios. In light of these biochemical processes, the  $\delta^{18}\text{O}_\text{P}$  values observed in soil samples have provided valuable information on the biological cycling of available P either in the natural environment (Angert et al., 2012; Tamburini et al., 2012; Gross et al., 2015) or in incubation studies using  $^{18}\text{O}$  labelled phosphate (Melby et al., 2013) or  $^{18}\text{O}$  labelled water (Gross and Angert, 2015). All of these studies demonstrated that labile soil P pools approach the expected oxygen isotopic equilibrium in soils. Yet, for more stable P pools, such as HCl P (a proxy of Ca-P), the changes in  $\delta^{18}\text{O}_\text{P}$  values upon labelling were generally small (Zohar et al., 2010; Joshi et al., 2016). It has been suggested that HCl P does not participate in biological P cycling but can incorporate the signal of biologically cycled P due to the continuous precipitation of secondary Ca-P minerals (Joshi et al., 2016). Thus, the  $\delta^{18}\text{O}_\text{P}$  signature of HCl P ( $\delta^{18}\text{O}_{\text{HCl P}}$ ) integrates long periods of time, which will cause a shift in  $\delta^{18}\text{O}_{\text{HCl P}}$  values but will not result in full equilibration. However, Amelung et al. (2015) reported that in an arable topsoil,  $\delta^{18}\text{O}_{\text{HCl P}}$  values were close to equilibrium and only with increasing depth,  $\delta^{18}\text{O}_{\text{HCl P}}$  values shifted progressively towards the isotopic signature of the parent material. Hence, in the long-term, even a putatively biologically inert P pool might in fact be part of the P cycle especially in the topsoil, possibly via the precipitation and dissolution of secondary minerals.

Similar effects might be observed for biopores, which over the course of decades integrate multiple generations of earthworms and roots and could thus enhance P cycling especially in the subsoil.

Therefore, we resampled the bulk soil and also biopores at the site described in Amelung et al. (2015) and analysed them by a combination of Hedley sequential extraction, XANES spectroscopy and  $\delta^{18}\text{O}_\text{P}$  of inorganic P extracted by 1 M HCl, in order to elucidate the effects of biopores on the biological turnover of the stable Ca-P pool. Further, we incubated bulk soil samples with  $^{18}\text{O}$ -labelled water and analysed the change in their  $\delta^{18}\text{O}_{\text{HCl P}}$  values to determine to which extent P cycling in the bulk subsoil affects the HCl P pool. We hypothesised that (1) due to P inputs by plant roots and especially earthworms, microbial P mobilisation and turnover in subsoil biopores is higher than in bulk soil indicated by greater concentrations of labile P in the Hedley sequential extraction and XANES spectroscopy and that (2) this higher cycling will shift  $\delta^{18}\text{O}_{\text{HCl P}}$  values of Ca-P closer to equilibrium values. However, we assume that (3) outside of biopores the microbial P turnover and subsequent precipitation as Ca-P is low, which should be reflected by  $\delta^{18}\text{O}_{\text{HCl P}}$  values further away from equilibrium and only slow changes in  $\delta^{18}\text{O}_{\text{HCl P}}$  values in the subsoil during the incubation.

## 2. Materials and methods

### 2.1. Site and soil sampling

Samples were collected from a haplic Luvisol at an eight-year field trial at the Research Station Campus Klein-Altendorf of the University of Bonn. The field trial was established to observe the effect of pre-cropping on the abundance of biopores in the soil as well as their utilisation by following cereal crops. The pre-crops under study were alfalfa (*Medicago sativa* L.), chicory (*Cichorium intybus* L.) and tall fescue (*Festuca arundinacea* Schreb.). For a more detailed description of the field trial see, e.g., Kautz et al. (2014) and a general characterisation of the soil is given by Vetterlein et al. (2013). The tall fescue treatments were previously sampled and analysed for its bulk soil  $\delta^{18}\text{O}_\text{P}$  signature in the study by Amelung et al. (2015). For the present study, we sampled three field replicates of the treatment with alfalfa (*Medicago sativa* L.) as a pre-crop for three years, followed by five years of a cereal-rapeseed rotation. The crop in the year of sampling was barley (*Hordeum vulgare* L.) and samples were collected in April, seven months after sowing. Biopores were identified as round voids in the soil (~5 mm in diameter) characterised by linings of a darker colour and in a few cases containing barley roots. Biopore material was collected by opening a biopore in 10 cm increments within specified depth intervals of 0–30, 30–45, 45–55, 55–75 and 75–105 cm depth. We sampled the linings of the biopores (~1 mm) until the bulk soil of a brighter colour appeared. All material from biopores was combined into a composite sample for each depth interval of each field replicate, respectively. As a reference, we also sampled bulk soil material excluding all areas around biopores. Samples were dried at 40 °C and sieved to 2 mm grain size.

### 2.2. General soil parameters

All samples were analysed for total elemental composition by microwave digestion in *aqua regia* and subsequent measurement using inductively coupled plasma-optical emission spectroscopy (ICP-OES; Thermo Fisher iCAP™ 7600). For the determination of the concentrations of total carbon (C), total nitrogen (N), and the C:N ratio we used dry combustion followed by the heat conductivity detection of the released trace gases (vario MICRO cube, Elementar, Hanau, Germany). For bulk soil samples we additionally checked for the presence of inorganic C by adding HCl to an aliquot of each sample and we determined pH ( $\text{CaCl}_2$ ) at a 1:2.5 soil:solution ratio; for biopore samples there were not sufficient sample amounts for these measurements. For bulk and biopore samples we

determined P pools by sequential extraction according to Hedley et al. (1982) as modified by Tiessen and Moir (1993). In deviation from these method scripts, we determined residual P by hot acid digestion (130 °C) in *aqua regia*. Inorganic P ( $P_i$ ) in the extracted solutions was determined photometrically by the molybdenum-blue method (Murphy and Riley, 1962), total P was again determined by ICP-OES and organic P ( $P_o$ ) was calculated as the difference of total and inorganic P.

### 2.3. XANES data collection and data treatment

The XANES spectra were recorded at the Soft X-Ray Micro-characterization Beamline (SXRMB) at the Canadian Light Source in Saskatoon, Saskatchewan, Canada, covering an energy range of 1.7–10 keV. An aliquot of the dried samples was ground to a fine powder using an agate stone mortar before being applied to the Cu sample holder by spreading the sample powder onto a double-sided carbon tape (G3939, Plano GmbH, Wetzlar, Germany). The spectra of the soil samples were recorded in fluorescence yield mode and the standard compound spectra were recorded in total electron yield mode at the P *K*-edge at photon energies between 2130 and 2220 eV. The step size was set to be 2 eV in the pre-edge range covering the energy range between 2130 and 2149 eV; 0.25 eV at the edge step covering the energy range between 2149 and 2172 eV, and 0.5 in the post-edge region covering an energy range between 2172 and 2220 eV. For all sample spectra a dwell time of 4 s was used and for the standard compound spectra a dwell time of 1 s was used. All samples were recorded three times and averaged to reduce the background. After each scan the sample holder position was slightly modified in order to prevent altering the sample due to the radiation.

The spectra averaging, background and baseline corrections of all XANES spectra were done using ATHENA (Ravel and Newville, 2005). All spectra were normalised to an edge-step of one before performing a Linear Combination Fitting (LCF) procedure. The LCF was carried out for all possible binary to quaternary combinations of all P reference compounds measured. The maximum of four P reference compounds was chosen to prevent an overfitting. No energy shift was allowed during the fit. The R-factor and the reduced chi-squared ( $\chi^2$ ) values were used as goodness-of-fit criteria and checked for plausibility and consistency with other obtained data (Kruse et al., 2015). For LCF the following P reference compounds were used:  $AlPO_4$ ,  $AlPO_4 \cdot 4H_2O$ ,  $FePO_4 \cdot 4H_2O$ ,  $MgHPO_4$ ,  $Mg_3(PO_4)_2 \cdot 8H_2O$ ,  $K_2HPO_4$ ,  $KH_2PO_4$ ,  $Ca(H_2PO_4)_2 \cdot xH_2O$ ,  $Ca_5(PO_4)_3(OH)$ ,  $NaH_2PO_4$ , O-phosphoryl ethanolamine, and phytic acid Na salt. The  $KH_2PO_4$  is our reference substance being unambiguously bioavailable, as it is easily soluble in water and thus the phosphate released is referred to as completely plant available (Audette et al., 2016). The other P species used are of lower bioavailability compared to  $KH_2PO_4$  but a ranking of P bioavailability always depends on soil properties. As the selection of critical reference compounds should be based on all available information of the sample (Kruse et al., 2015), we based the selection here on total elemental composition as well as the results of the Hedley sequential extraction of the samples. In addition, P reference compounds were added that typically are present in the soil (Kruse et al., 2015). All reference compounds were purchased from Sigma Aldrich in reagent grade and were treated similar to the soil samples.

### 2.4. Soil incubation

In order to determine P turnover at different depths, we incubated bulk soil samples as a composite of the three field replicates for the depth intervals of 0–30, 45–55 and 75–105 cm. A similar

incubation of biopore material was not possible due to a lack of sufficient sample amounts. For each of the bulk soil samples we determined the water holding capacity (WHC) of the soil according to Foster (1995). As total drying of the samples might affect microbial activity, samples were only air-dried at 4 °C to 25% of WHC. We prepared water labelled to  $\delta^{18}O_W = 1949\text{‰}$  by diluting highly labelled water (97%  $^{18}O$ ; Euriso-Top GmbH, Saarbrücken, Germany). We used this labelled water for rewetting the samples to 50% of WHC, which resulted in a total  $\delta^{18}O_W \approx 978\text{‰}$  assuming that the water still remaining in the sample had  $\delta^{18}O_W \approx -8\text{‰}$  (based on soil water  $\delta^{18}O_W$  signatures estimated by Amelung et al., 2015). Samples were incubated in glass vials at room temperature ( $21 \pm 1$  °C). Twice a week the glass vials were opened to replenish oxygen and to check water content by weighing. Within these intervals, only minimal amounts of water had evaporated (<0.5 mL), which were then replaced with labelled water. For each depth, we prepared six vials, half of which were incubated for one week, while the other half was incubated for four weeks. After the respective incubation interval, samples were dried at 40 °C.

### 2.5. Oxygen isotopic composition of phosphate ( $\delta^{18}O_{HCl P}$ )

In preparation of the analysis of the oxygen isotopic composition of HCl-extractable phosphate ( $\delta^{18}O_{HCl P}$ ) all dried soil samples were sequentially extracted with 0.5 M  $NaHCO_3$ , 0.1 M NaOH and 1 M HCl at a 1:10 soil:solution ratio (Amelung et al., 2015). The alkaline extracts were not analysed for  $\delta^{18}O_P$  here, but they removed most of the organic compounds and polyphosphates from the samples. The HCl-extract was then almost free of organic contaminations, which is essential for the subsequent purification of phosphate from the soil solution. For the purification of the HCl-extracts, we followed the procedure by Tamburini et al. (2010) with sequential precipitation, washing and dissolution of phosphate as ammonium-phosphomolybdate and magnesium-ammonium-phosphate, followed by rinsing of the solute on a cation exchange resin and a final precipitation of phosphate as silver phosphate ( $Ag_3PO_4$ ).  $Ag_3PO_4$  crystals were dried at 50 °C and subsequently stored in a desiccator until analysis.  $Ag_3PO_4$  was analysed for  $\delta^{18}O_P$  by the Group of Plant Nutrition at ETH Zurich using a vario PYRO Cube (Elementar, Hanau, Germany) with a carbon-based reactor connected in continuous flow to an Isoprime 100 isotope ratio mass spectrometer (Isoprime, Manchester, UK). Furnace temperature of the reactor was set to 1450 °C and resulting CO and N<sub>2</sub> gases were separated by a purge and trap chromatography system. Results were calibrated against an internal  $Ag_3PO_4$  standard (Acros Organics, Geel, Belgium;  $\delta^{18}O = 14.2\text{‰}$ ) and two benzoic acid standards (International Atomic Energy Agency, Vienna, Austria; IAEA 601:  $\delta^{18}O = 23.1\text{‰}$  and IAEA 602:  $\delta^{18}O = 71.3\text{‰}$ ). All values are reported in the conventional delta notation and are calibrated against the Vienna Standard Mean Oceanic Water (VSMOW).

Theoretical equilibrium values for field samples were calculated based on the equation of Chang and Blake (2015) (Eq. (1)):

$$\delta^{18}O_P = e^{\left(\frac{14.43 - 26.54}{T}\right)} \times (\delta^{18}O_W + 1000) - 1000 \quad (1)$$

where  $T$  is the ambient soil temperature in [K] and  $\delta^{18}O_P$  and  $\delta^{18}O_W$  are the  $^{18}O:^{16}O$  ratios of phosphate and water, respectively. For the field samples, based on data given by Amelung et al. (2015), we assumed minimum, maximum, and mean temperatures of 3.8 °C, 11.8 °C and 6.8 °C, respectively, and  $\delta^{18}O_W$  values of  $-8.4\text{‰}$ . We did not measure  $\delta^{18}O_W$  of the samples, because we preferred annual mean  $\delta^{18}O_W$  values over  $\delta^{18}O_W$  values of the day of sampling, since  $\delta^{18}O_{HCl P}$  integrates the isotopic signal over long periods of time

(Angert et al., 2012; Joshi et al., 2016) and is therefore likely not related to soil  $\delta^{18}\text{O}_\text{W}$  values on one specific day. Equilibrium values for the incubated samples were calculated in the same way using the incubation temperature of 21 °C and  $\delta^{18}\text{O}_\text{W}$  of 978‰.

## 2.6. Statistics

Statistical analyses were performed in SigmaPlot 13.0 (Systat Software Inc. USA). We tested for normal distribution by Shapiro-Wilk test ( $P < 0.05$ ) and for homogeneity of variances by Brown-Forsythe test ( $P < 0.05$ ). We considered both bulk soil and biopore samples within each depth as well as samples from different depths within one soil profile as paired samples. To account for this, we performed separate Repeated Measures ANOVAs, where depth intervals either in bulk soil or in biopores were treated as repeated measures. If significant differences occurred we used the Tukey HSD test for a post-hoc separation of means ( $P < 0.05$ ). To test for the effect of bulk vs. biopores, we performed paired t-tests for each depth interval separately ( $P < 0.05$ ). For incubated samples, we performed a Repeated Measures ANOVA with Tukey HSD post-hoc test ( $P < 0.05$ ) for each incubated depth separately. As the data for topsoil samples were not normally distributed, we used a One-Way Repeated Measures ANOVA on Ranks for this sample set.

## 3. Results

### 3.1. General soil parameters and P pools

Biopores were characterised by higher nutrient concentrations for all elements except magnesium (Mg) (Table 1). However, in most cases these effects were not significant at  $P < 0.05$ , which is probably due to the small number of replicates ( $n = 3$ ). In the topsoil (0–30 cm) differences in nutrients concentrations between bulk and biopore were small, but increased with depth. The pH value could only be assessed for the bulk soil, due to limited sample amounts for the biopores. Results showed that pH values increased significantly from slightly acidic conditions in the topsoil towards

neutrality at depth ( $P < 0.05$ ). We did not observe inorganic carbon in any of the bulk soil samples.

Total P concentrations were highest in the topsoil and were substantially lower at intermediate depths (30–45 cm and 45–55 cm) but increased again below 55 cm. Considering specific P pools as determined by Hedley sequential fractionation, P concentrations in biopores were mostly greater than in the bulk soil, although this effect was significant in only few cases. The increase in P concentrations in individual P pools matched the gain in total P. Thus, the relative distribution of P among Hedley P pools was more or less similar for bulk soil and biopores, with the latter exhibiting only a slightly but not significantly greater proportion of labile P (i.e. resin P and  $\text{NaHCO}_3$  P; Table 2).

### 3.2. P K-edge XANES

First visual analysis of the soil XANES spectra (exemplary topsoil spectra are shown in Fig. 1) revealed the main signal (white line) at 2152.3 electron-volts (eV). A small pre-edge feature around 2146 eV was visible in all samples. This pre-edge feature hints at the presence of Fe-P, which was to be expected considering the high concentrations of Fe determined by acid digestion. However, visual inspection alone does not sufficiently provide evidence of P speciation, therefore we have performed an LCF procedure.

The LCF indicated  $\text{FePO}_4 \times 4\text{H}_2\text{O}$ ,  $\text{KH}_2\text{PO}_4$ , and  $\text{AlPO}_4$  as the predominant P species in all samples. For bulk soil samples, the proportions of  $\text{AlPO}_4$  and  $\text{FePO}_4 \times 4\text{H}_2\text{O}$  increased with depth, while  $\text{KH}_2\text{PO}_4$  and  $\text{Mg}_3(\text{PO}_4)_2 \times 8\text{H}_2\text{O}$  decreased (Table 3). This is in line with elemental concentrations of Fe and Mg as determined by acid digestion (Table 1). However, in the Hedley sequential extraction, concentrations of  $\text{NaHCO}_3$  P and NaOH P, which are assumed to extract P bound to Fe- and Al-oxides, decreased with depth (Table 2). In biopore samples, no Mg-P was detected by LCF (Table 3), similar to the depletion in total Mg in biopores relative to bulk soil (Table 1). In addition,  $\text{Ca}(\text{H}_2\text{PO}_4)_2 \times \text{H}_2\text{O}$  could be quantified in the subsoil biopores, whereas it was not present in all other samples, in line with the higher total Ca concentrations in subsoil

**Table 1**  
Elemental composition of soil samples [ $\text{g kg}^{-1}$ ] given as mean and standard error [SE] ( $n = 3$ ). C and N determined by dry total combustion, all other elements determined by microwave digestion in *aqua regia*, pH determined in  $\text{CaCl}_2$  solution. Different lowercase letters indicate significant differences with depth for bulk soil or biopore samples, respectively, whereas different uppercase letters indicate significant differences of biopores and bulk soil at the respective depth ( $P < 0.05$ ).

| Sample         |      | C [ $\text{g kg}^{-1}$ ] |    | N [ $\text{g kg}^{-1}$ ] |       | C/N [–] |       | Ca [ $\text{g kg}^{-1}$ ] |       | Fe [ $\text{g kg}^{-1}$ ] |    | Mn [ $\text{g kg}^{-1}$ ] |    | Mg [ $\text{g kg}^{-1}$ ] |    | S [ $\text{g kg}^{-1}$ ] |    | pH [–]            |    |
|----------------|------|--------------------------|----|--------------------------|-------|---------|-------|---------------------------|-------|---------------------------|----|---------------------------|----|---------------------------|----|--------------------------|----|-------------------|----|
| <b>Bulk</b>    |      |                          |    |                          |       |         |       |                           |       |                           |    |                           |    |                           |    |                          |    |                   |    |
| 0–30           | mean | 9.6                      | a  | A                        | 1.1   | a       | 9.2   | a                         | 4.1   | 21                        | a  | 0.82                      | a  | 3.2                       | a  | 0.15                     | a  | 6.7               | a  |
|                | (SE) | (0.4)                    |    |                          | (0.1) |         | (0.5) |                           | (0.3) | (1)                       |    | (0.01)                    |    | (0.2)                     |    | (0)                      |    | (0)               |    |
| 30–45          | mean | 5.3                      | b  |                          | 0.7   | b       | 8.0   | b                         | 3.6   | 23                        | a  | 0.78                      | ab | 3.6                       | ab | 0.09                     | b  | 6.8               | ab |
|                | (SE) | (0.6)                    |    |                          | (0.1) |         | (0.1) |                           | (0.3) | (1)                       |    | (0.02)                    |    | (0.1)                     |    | (0)                      |    | (0)               |    |
| 45–55          | mean | 4.2                      | bc |                          | 0.6   | bc      | 7.2   | bc                        | 4.3   | 30                        | b  | 0.67                      | ab | 4.4                       | bc | 0.09                     | b  | 6.8               | b  |
|                | (SE) | (0.2)                    |    |                          | (0)   |         | (0.2) |                           | (0)   | (3)                       |    | (0.07)                    |    | (0.1)                     |    | (0.01)                   |    | (0)               |    |
| 55–75          | mean | 3.9                      | bc |                          | 0.5   | bc      | 6.9   | c                         | 4.4   | 31                        | b  | 0.62                      | b  | 5.0                       | cd | 0.07                     | b  | 6.9               | bc |
|                | (SE) | (0.1)                    |    |                          | (0)   |         | (0)   |                           | (0.1) | (1)                       |    | (0.01)                    |    | (0.3)                     |    | (0)                      |    | (0)               |    |
| 75–105         | mean | 2.8                      | c  |                          | 0.4   | c       | 6.2   | d                         | 4.7   | 30                        | b  | 0.74                      | ab | 5.2                       | d  | 0.07                     | b  | 7.0               | c  |
|                | (SE) | (0.2)                    |    |                          | (0)   |         | (0.2) |                           | (0.2) | (1)                       |    | (0.05)                    |    | (0.2)                     |    | (0)                      |    | (0.1)             |    |
| <b>Biopore</b> |      |                          |    |                          |       |         |       |                           |       |                           |    |                           |    |                           |    |                          |    |                   |    |
| 0–30           | mean | 12.1                     |    | B                        | 1.2   | a       | 9.7   |                           | 4.3   | 21                        | a  | 0.83                      |    | 3.1                       |    | 0.18                     | a  | n.a. <sup>a</sup> |    |
|                | (SE) | (0.1)                    |    |                          | (0)   |         | (0.2) |                           | (0.2) | (1)                       |    | (0.01)                    |    | (0.3)                     |    | (0.02)                   |    |                   |    |
| 30–45          | mean | 8.9                      |    |                          | 1.0   | ab      | 8.8   |                           | 4.7   | 23                        | ab | 0.82                      |    | 3.7                       |    | 0.14                     | ab | n.a.              |    |
|                | (SE) | (0.9)                    |    |                          | (0.1) |         | (0.3) |                           | (0.2) | (1)                       |    | (0.01)                    |    | (0.1)                     |    | (0.01)                   |    |                   |    |
| 45–55          | mean | 9.3                      |    |                          | 0.9   | ab      | 8.4   |                           | 5.6   | 24                        | ab | 0.73                      |    | 3.9                       |    | 0.13                     | b  | n.a.              |    |
|                | (SE) | (1.8)                    |    |                          | (0.1) |         | (0.2) |                           | (0.1) | (1)                       |    | (0.01)                    |    | (0.2)                     |    | (0.01)                   |    |                   |    |
| 55–75          | mean | 10.3                     |    |                          | 1.0   | ab      | 10    |                           | 7.2   | 25                        | ab | 0.72                      |    | 4.2                       |    | 0.14                     | ab | n.a.              |    |
|                | (SE) | (1.8)                    |    |                          | (0.1) |         | (0.9) |                           | (0.8) | (1)                       |    | (0.04)                    |    | (0.3)                     |    | (0.01)                   |    |                   |    |
| 75–105         | mean | 7.2                      |    |                          | 0.8   | b       | 9.0   |                           | 7.0   | 27                        | b  | 0.79                      |    | 4.6                       |    | 0.11                     | b  | n.a.              |    |
|                | (SE) | (0.1)                    |    |                          | (0)   |         | (0.1) |                           | (0.7) | (0)                       |    | (0.05)                    |    | (0.1)                     |    | (0.01)                   |    |                   |    |

<sup>a</sup> n. a. = measurements were not possible due to lack sufficient amounts of sample material.

**Table 2**

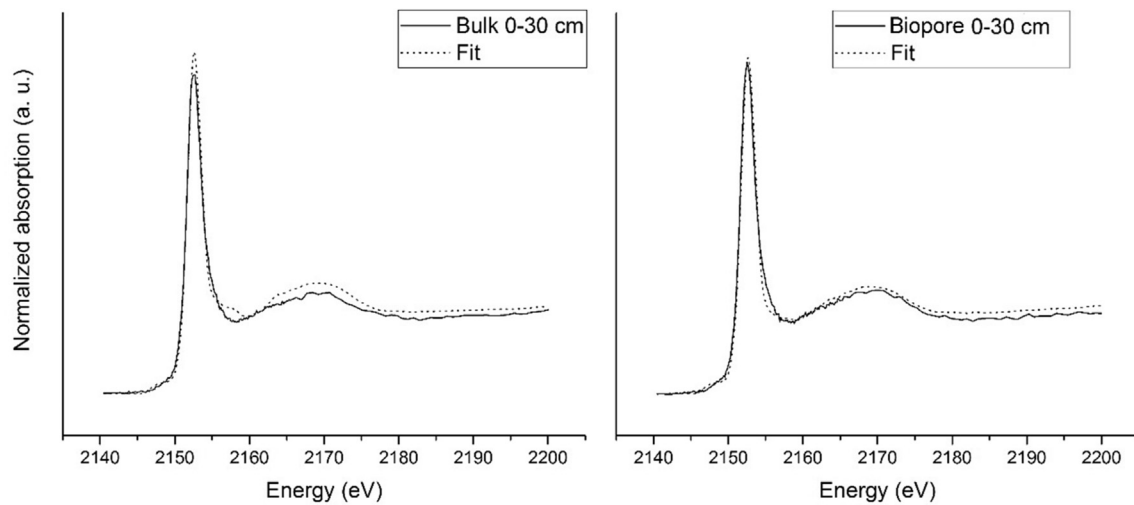
Concentrations of P pools determined by sequential extraction (Hedley) and total P concentrations determined by microwave digestion in *aqua regia* given as mean and standard error [SE] (n = 3). Different lowercase letters indicate significant differences with depth for bulk soil or biopore samples, respectively, whereas different uppercase letters indicate significant differences of biopores and bulk soil at the respective depth ( $P < 0.05$ ).

| Sample         | Total P |                        | Resin P                |              | NaHCO <sub>3</sub> P <sub>i</sub> |              | NaHCO <sub>3</sub> P <sub>o</sub> |              | NaOH P <sub>i</sub>    |              | NaOH P <sub>o</sub>    |              | HCl P                  |              |                  |      |     |      |      |      |      |
|----------------|---------|------------------------|------------------------|--------------|-----------------------------------|--------------|-----------------------------------|--------------|------------------------|--------------|------------------------|--------------|------------------------|--------------|------------------|------|-----|------|------|------|------|
|                |         | [mg kg <sup>-1</sup> ] | [mg kg <sup>-1</sup> ] | % of total P | [mg kg <sup>-1</sup> ]            | % of total P | [mg kg <sup>-1</sup> ]            | % of total P | [mg kg <sup>-1</sup> ] | % of total P | [mg kg <sup>-1</sup> ] | % of total P | [mg kg <sup>-1</sup> ] | % of total P |                  |      |     |      |      |      |      |
| <b>Bulk</b>    |         |                        |                        |              |                                   |              |                                   |              |                        |              |                        |              |                        |              |                  |      |     |      |      |      |      |
| 0–30           | mean    | 702 <sup>a</sup>       | 90                     | a            | 12.9                              | 45           | a                                 | 7.1          | 2                      | 0.3          | 81                     | a            | 11.6                   | 26           | 3.8              | 228  | a   | 32.5 |      |      |      |
|                | (SE)    | (90)                   | (4)                    |              |                                   | (0)          |                                   |              | (1)                    |              | (2)                    |              |                        | (6)          |                  | (19) |     |      |      |      |      |
| 30–45          | mean    | 471                    | b                      | 33           | b                                 | A            | 7.1                               | 31           | b                      | 6.6          | n.d. <sup>b</sup>      |              | 58                     | b            | 12.2             | 11   | 2.4 | 131  | b    | A    | 27.8 |
|                | (SE)    | (33)                   | (6)                    |              |                                   | (4)          |                                   |              |                        |              |                        |              | (7)                    |              | (–) <sup>c</sup> |      |     | (2)  |      |      |      |
| 45–55          | mean    | 440                    | b                      | 19           | b                                 | 4.3          | 25                                | b            | 5.7                    | n.d.         |                        | 50           | b                      | 11.4         | 19               | 4.4  | 116 | b    | 26.4 |      |      |
|                | (SE)    | (19)                   | (4)                    |              |                                   | (4)          |                                   |              |                        |              |                        |              | (4)                    |              | (–)              |      |     | (4)  |      |      |      |
| 55–75          | mean    | 504                    | b                      | 16           | b                                 | 3.2          | 26                                | b            | 5.2                    | n.d.         |                        | 52           | b                      | A            | 10.3             | 7    | 1.3 | 168  | b    | 33.3 |      |
|                | (SE)    | (16)                   | (5)                    |              |                                   | (3)          |                                   |              |                        |              |                        |              | (2)                    |              | (–)              |      |     | (19) |      |      |      |
| 75–105         | mean    | 609 <sup>a</sup>       | a                      | 16           | a                                 | 2.6          | 24                                | b            | 4.0                    | n.d.         |                        | 49           | b                      | A            | 8.0              | 7    | 1.1 | 279  | a    | 45.8 |      |
|                | (SE)    | (16)                   | (6)                    |              |                                   | (4)          |                                   |              |                        |              |                        |              | (5)                    |              | (3)              |      |     | (3)  |      |      |      |
| <b>Biopore</b> |         |                        |                        |              |                                   |              |                                   |              |                        |              |                        |              |                        |              |                  |      |     |      |      |      |      |
| 0–30           | mean    | 686 <sup>a</sup>       | a                      | 93           | ab                                | 13.5         | 49                                | a            | 7.2                    | 15           | 2.2                    | 75           |                        | 10.9         | 30               | 4.3  | 197 | ab   | 27.8 |      |      |
|                | (SE)    | (93)                   | (–)                    |              |                                   | (–)          |                                   |              |                        | (–)          |                        | (–)          |                        |              | (–)              |      |     | (–)  |      |      |      |
| 30–45          | mean    | 606                    | ab                     | 46           | ab                                | B            | 7.5                               | 39           | ab                     | 6.4          | n.d.                   |              | 67                     |              | 11.0             | 11   | 1.8 | 177  | ab   | B    | 29.2 |
|                | (SE)    | (46)                   | (3)                    |              |                                   | (2)          |                                   |              |                        |              |                        |              | (1)                    |              |                  | (–)  |     | (4)  |      |      |      |
| 45–55          | mean    | 550                    | b                      | 31           | b                                 | 5.6          | 33                                | b            | 5.9                    | n.d.         |                        | 65           |                        | 11.8         | 10               | 1.8  | 156 | a    | 28.3 |      |      |
|                | (SE)    | (31)                   | (3)                    |              |                                   | (1)          |                                   |              |                        |              |                        |              | (7)                    |              | (5)              |      |     | (10) |      |      |      |
| 55–75          | mean    | 559                    | ab                     | 25           | ab                                | 4.4          | 33                                | b            | 5.9                    | n.d.         |                        | 61           | B                      | 10.9         | 26               | 4.6  | 207 | ab   | 37.1 |      |      |
|                | (SE)    | (25)                   | (0)                    |              |                                   | (2)          |                                   |              |                        |              |                        |              | (3)                    |              | (–)              |      |     | (27) |      |      |      |
| 75–105         | mean    | 627 <sup>a</sup>       | ab                     | 25           | a                                 | 4.0          | 32                                | b            | 5.1                    | 1            | 0.2                    | 56           | B                      | 8.9          | 0                | 0    | 239 | b    | 38.1 |      |      |
|                | (SE)    | (25)                   | (4)                    |              |                                   | (1)          |                                   |              |                        | (0)          |                        |              | (4)                    |              | (–)              |      |     | (14) |      |      |      |

<sup>a</sup> Data from Hoang et al. (under review).

<sup>b</sup> n. d. = not detected.

<sup>c</sup> Standard error (–) indicates that data for only one field replicate were available, for these data a paired *t*-test for the comparison of bulk and biopore samples was not possible.



**Fig. 1.** Normalised P K-edge XANES spectra of the samples at a depth of 0–30 cm (solid line) and the corresponding fit (dashed line).

**Table 3**

Proportions of P species [%] as obtained by deconvolution of the P K-edge XANES spectra of the soil samples by using linear combination fitting (LCF).

| Sample         | AlPO <sub>4</sub> | FePO <sub>4</sub> ·4H <sub>2</sub> O | Mg <sub>3</sub> (PO <sub>4</sub> ) <sub>2</sub> · 8H <sub>2</sub> O | KH <sub>2</sub> PO <sub>4</sub> | Organically bound P <sup>a</sup> | Ca(H <sub>2</sub> PO <sub>4</sub> ) <sub>2</sub> ·xH <sub>2</sub> O | R-factor | χ <sup>2</sup> |
|----------------|-------------------|--------------------------------------|---------------------------------------------------------------------|---------------------------------|----------------------------------|---------------------------------------------------------------------|----------|----------------|
| <b>Bulk</b>    |                   |                                      |                                                                     |                                 |                                  |                                                                     |          |                |
| 0–30           | 21.8              | 29.3                                 | 19.5                                                                | 29.4                            | 0                                | 0                                                                   | 0.0038   | 1.07           |
| 30–45          | 26.1              | 35.6                                 | 14.0                                                                | 24.3                            | 0                                | 0                                                                   | 0.0025   | 0.76           |
| 75–105         | 25.5              | 34.9                                 | 12.8                                                                | 26.8                            | 0                                | 0                                                                   | 0.0027   | 0.78           |
| <b>Biopore</b> |                   |                                      |                                                                     |                                 |                                  |                                                                     |          |                |
| 0–30           | 22.1              | 32.2                                 | 0                                                                   | 10.3                            | 35.4                             | 0                                                                   | 0.0018   | 0.54           |
| 30–45          | 16.1              | 24.9                                 | 0                                                                   | 33.6                            | 0                                | 25.4                                                                | 0.0035   | 1.1            |
| 75–105         | 7.1               | 31.1                                 | 0                                                                   | 38.4                            | 0                                | 23.4                                                                | 0.0034   | 1.1            |

<sup>a</sup> Phytic acid sodium salt hydrate.

biopores (Table 1). Finally, LCF indicated the presence of phytic acid sodium hydrate salt as a proxy for organically bound P in topsoil biopore samples (Table 3). This also matched the results of the Hedley sequential extraction, where topsoil biopores had the highest proportion of  $P_o$  compared to subsoil or bulk soil samples (Table 2).

### 3.3. $\delta^{18}O_{HCl\ P}$ values

$\delta^{18}O_{HCl\ P}$  values decreased with depth in bulk soil and, in tendency, also in biopore samples (Fig. 2). Even though biopore  $\delta^{18}O_{HCl\ P}$  values appeared to be generally greater than for the bulk soil, the difference in  $\delta^{18}O_{HCl\ P}$  values was not statistically significant at  $P < 0.05$ . Below 55 cm soil depth,  $\delta^{18}O_{HCl\ P}$  in bulk soil remained constant at approximately 13‰, whereas biopore  $\delta^{18}O_{HCl\ P}$  values were greater and were also still close to equilibrium values even in the deeper subsoil.

The incubation of bulk soil samples without biopores at room temperature with  $^{18}O$ -labelled water did not result in a significant increase in HCl P concentrations over time ( $P < 0.05$ ). Nevertheless, there was an increase in topsoil  $\delta^{18}O_{HCl\ P}$  values from 16‰ to 21‰ already after one week and to 31‰ after four weeks, which represented a significant increase compared to the initial  $\delta^{18}O_{HCl\ P}$  values. Nevertheless, even topsoil  $\delta^{18}O_{HCl\ P}$  values at the end of the incubation were still far from the expected theoretical equilibrium value of ~1000‰.

In the subsoil, by contrast,  $\delta^{18}O_{HCl\ P}$  values changed rather slowly if at all (not significant).

## 4. Discussion

The scale of nutrient accumulation in biopores might depend on whether pores were created and inhabited by roots or by earthworms. For the pores sampled in this study, no specific information on pore history was available. However, Kautz et al. (2014) demonstrated that perennial fodder cropping affected the formation of new biopores at the test site more than the activity of earthworms. Also, Athmann et al. (2013) reported that up to 80% of biopores at the site contained roots. Nevertheless, we assume that the pores sampled in this study incorporated effects of both

earthworms and roots and we therefore will not attempt to separate these two processes.

Biopores in our study were characterised by higher nutrient concentrations than bulk soil from the same depth (Table 1). This is in agreement with a number of previous studies reporting an increase in nutrient concentrations (Tiunov and Scheu, 1999; Don et al., 2008). However, we also found these differences to be less pronounced in the topsoil (0–30 cm), which can be attributed to the replenishment of nutrients by inputs of organic matter as well as the frequent mixing of the soil by tillage operations. Therefore, the beneficial effect of biopores in the topsoil is relatively small due to the already high concentration of nutrients. By comparison, in the subsoil with lower nutrient concentrations, the relative increase in nutrient concentration in biopores is much more pronounced.

Considering P speciation in biopores, XANES spectroscopy indicated that organic P ( $P_o$ ) was present in topsoil biopore samples (Table 3). Yet more reliable data on  $P_o$  forms are obtained with sequential Hedley extraction, which confirmed large proportions but no preferential enrichment of  $P_o$  in biopores in the topsoil (Table 2). For all other biopore samples and also for the bulk soil, XANES spectroscopy did not indicate the presence of phytic acid as a proxy for  $P_o$  (Table 3). It has repeatedly been demonstrated that the identification of  $P_o$  compounds by XANES spectroscopy is complicated by interferences, the lack of specific spectral features and low  $P_o$  concentrations (Beauchemin et al., 2003; Kruse et al., 2015). In a study on arable soils Eriksson et al. (2015) were able to detect  $P_o$  by XANES spectroscopy only in soils with organic C concentrations > 2%. By comparison, total C concentration in topsoil biopores was 1.2% and thus probably already close to the detection limit, which would account for failure to detect  $P_o$  in other samples with even lower C concentrations.

Among the inorganic P ( $P_i$ ) species considered in the LCF analysis, Mg-P was present only in the bulk soil samples, but not in biopore samples (Table 3). This was in line with low concentrations of Mg in biopores (Table 1), possibly because Mg is easily soluble and therefore rather mobile in soils (Gransee and Führs, 2013), i.e., it is more prone to leaching in biopores, which present a preferential flow path for soil water (Beven and Germann, 1982; Hagedorn and Bundt, 2002). Further, the contribution of Fe- and Al-P in biopores decreased with depth from 54% to 38%, mostly due to a marked decrease in Al-P (Table 3), whereas for the bulk soil, Fe- and Al-P contributed 50% of total P in the topsoil and 60% in the subsoil. This increase in the relative contribution of Fe- and Al-P in bulk soil was not supported by the results of the Hedley sequential extraction, where concentrations of  $NaHCO_3$ -P and  $NaOH$ -P as well as their relative contribution to total P decreased with depth. Here it should be noted that the distinction of various Fe-bonding forms by XANES spectroscopy is rather difficult due to the complex soil matrix, and thus XANES might include more stable Fe-P compounds, which are not extracted by  $NaHCO_3$  and  $NaOH$ , but can instead be included in the HCl-extract (Slomp et al., 1996).

In subsoil biopores, LCF of the XANES spectra quantified higher proportions of the easily soluble and highly bioavailable  $KH_2PO_4$  in subsoil biopores hinting at a higher bioavailability of P (Table 3). This finding matches the trend for higher concentrations of labile P in biopores as observed in the Hedley sequential extraction (Table 2). The accumulation of P in plant-available forms in biopores has been observed before (Kuczak et al., 2006; Barej et al., 2014). However, as  $KH_2PO_4$  is more soluble compared to, e.g., Mg-P, it is also less likely to persist in soils at high quantities (Singh and Gräfe, 2010). Also, the interpretation of  $KH_2PO_4$  in XANES spectra is complicated by the fact that both K-P and Ca-P spectra have a post white line shoulder (Singh and Gräfe, 2010). Thus, the increase in the K-P signals in subsoil biopores could also be an

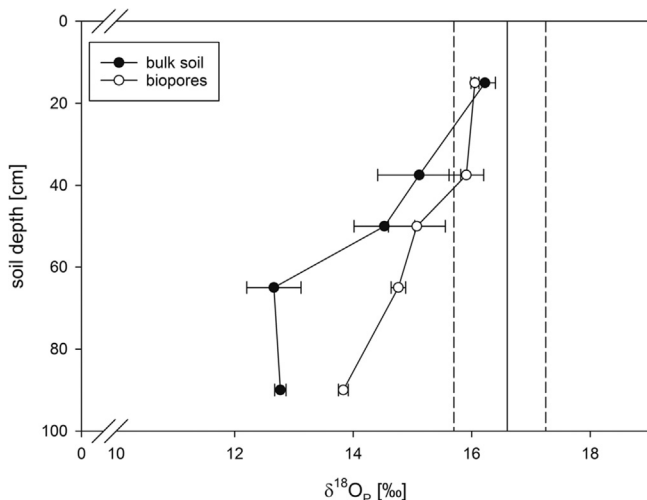


Fig. 2.  $\delta^{18}O_P$  profile in bulk soil and biopores, respectively and the theoretical equilibrium  $\delta^{18}O_P$  values at the site calculated with the annual minimum and maximum temperatures (dotted vertical lines) as well as annual mean temperatures (solid vertical line) as reported in Amelung et al. (2015). Error bars represent standard error of the mean ( $n = 3$ ).

artefact of higher Ca-P concentrations, as indicated by the detection of Ca-P itself in subsoil biopore samples (Table 3).

The occurrence of Ca-P in subsoil biopores could indicate a higher pH compared to bulk soil. However, even in the slightly acidic to neutral conditions (pH > 6) measured in bulk soil samples the occurrence of Ca-P minerals is possible (Hinsinger, 2001) and also Ca-P has also been detected by XANES spectroscopy in acidic soils (Beauchemin et al., 2003). Further, along with the increased total Ca concentrations (Table 1), the occurrence of Ca-P implies an additional input of Ca into the biopores. This could be attributed to the activity of earthworms, as it has been observed that earthworms excrete Ca-carbonate particles (Canti and Pearce, 2003; Lee et al., 2008). Yet, this finding is only partly in agreement with the results for HCl P of the Hedley sequential extraction, which is assumed to represent mainly Ca-P (Hedley et al., 1982). Here, the concentrations do not significantly differ between biopores and bulk soil of the deeper subsoil (Table 2). To better understand the underlying mechanisms as well as their implications for biological P cycling in the subsoil, we analysed the oxygen isotopic signature of HCl P ( $\delta^{18}\text{O}_{\text{HCl P}}$ ).

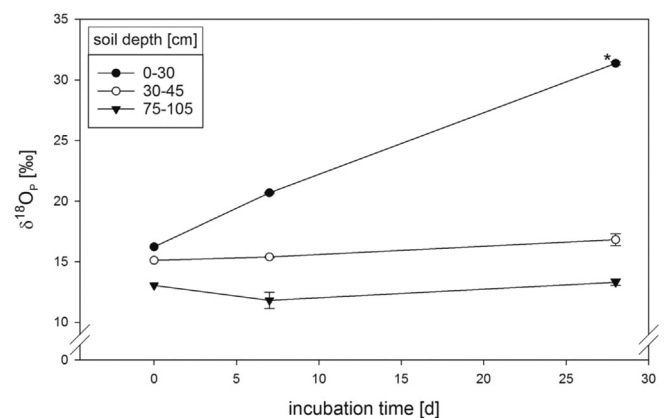
The depth profile of bulk soil  $\delta^{18}\text{O}_{\text{HCl P}}$  values was similar to the one reported for the same site by Amelung et al. (2015), although the isotopic signatures in the topsoil were slightly lower than those reported by Amelung et al. (2015) but nevertheless close to equilibrium. The lower topsoil values in our study might be an artefact of the sampling time, as our samples were taken in spring. The samples for the study by Amelung et al. (2015) by contrast were taken in late autumn and thus probably also include the higher isotopic signatures of phosphates equilibrated with soil water that had been enriched by evaporation during the summer months. In the subsoil  $\delta^{18}\text{O}_{\text{P}}$  values approached those expected for the calcareous parent material of ~10‰ in the deeper subsoil (Amelung et al., 2015). In biopores, by contrast,  $\delta^{18}\text{O}_{\text{HCl P}}$  values in the subsoil were greater than those in the bulk soil, i.e., they were closer to equilibrium (Fig. 2). A possible explanation here can be found in the fact that anecic earthworms feed on organic matter from the soil surface (Tiunov and Scheu, 1999; Don et al., 2008). As conditions in the earthworm gut are favourable to sustain high levels of enzyme activity,  $\text{P}_i$  will be mineralised from the ingested organic matter and subsequently excreted in labile forms into the soil (Kuczak et al., 2006; Vos et al., 2014). Further, the high level of enzyme activity in the earthworm gut will also induce intensive P cycling even before the excretion. Therefore, P imported into biopores in this way should have a  $\delta^{18}\text{O}_{\text{P}}$  value closer to equilibrium. In our study, this input of labile  $\text{P}_i$  into subsoil biopores is evident in the tendency for increased concentrations of resin P and in the basic extracts (Table 2) as well as higher proportions of the easily soluble  $\text{KH}_2\text{PO}_4$  estimated by XANES spectroscopy (Table 3). Once introduced into the biopores, P will further be equilibrated by enzymes derived from earthworm casts as well as from microorganisms of the rhizosphere of roots growing through the pore. We did not analyse the labile P pools for their  $\delta^{18}\text{O}_{\text{P}}$  signature to directly verify this process. However, the precipitation of P from these labile pools will incorporate the isotopic signature of biologically cycled P into secondary Ca-P minerals (Joshi et al., 2016). Also, as biopores are stable over the course of decades (Beven and Germann, 1982; Hagedorn and Bundt, 2002) this process will be repeated several times, resulting in an integrated signal in the  $\delta^{18}\text{O}_{\text{HCl P}}$  signature. The higher  $\delta^{18}\text{O}_{\text{HCl P}}$  values in subsoil biopores thus most likely represent the continuous and repeated input of labile, previously biologically cycled P.

In the deeper subsoil, the formation of secondary Ca-P minerals in biopores might be further promoted by the input of Ca-particles by earthworms (Canti and Pearce, 2003; Lee et al., 2008), as already discussed above for the results of XANES spectroscopy.

Thus, in the bulk soil, the incorporation of labile P into the HCl P pool and the resulting increase in  $\delta^{18}\text{O}_{\text{HCl P}}$  values is not only limited by a lack of P inputs and related stimulations of enzymatic activity in itself, but it is further limited by the low levels of Ca and missing additional inputs of Ca by earthworms.

Altogether, the higher concentrations of labile P apparently should render biopores into hotspots for P acquisition in the subsoil. However, the gradual departure of biopore  $\delta^{18}\text{O}_{\text{P}}$  values from the equilibrium value could indicate a decreasing influence of enzymatic P cycling with depth, probably due to less frequent visits of earthworms and less frequent rooting in the deeper parts of the biopores, which will reduce the abundance and activity of microorganisms and associated enzymes introduced with earthworm casts and the rhizosphere. Also, the very localised effect limits the overall impact on plant P supply. Barej et al. (2014) show that total P stocks in biopores at our field site were only 1–1.5% relative to those of the bulk soil. Thus, a by far greater effect can be generated if biopores promote plant access and utilisation of the vast amounts of P present in the bulk subsoil (Kautz et al., 2013). It is unclear though, which time scale is necessary to visualise such effects in the  $\delta^{18}\text{O}_{\text{HCl P}}$  signature.

In our study, we observed a change in topsoil  $\delta^{18}\text{O}_{\text{HCl P}}$  values of Ca-P by 15‰ within four weeks of incubation with  $^{18}\text{O}$  labelled water (Fig. 3), which can again result from the precipitation of labile P into the Ca-P pool. However, even if this is only a small increase by 1.5% relative to the theoretical value expected for full equilibration, this effect was unexpectedly strong, considering the suggested biologically inert behaviour of this pool (Joshi et al., 2016). Intriguingly, the change in  $\delta^{18}\text{O}_{\text{HCl P}}$  values was not accompanied by an increase in HCl P concentrations as would be expected if changes were caused only by precipitation without re-cycling. Similar observations in other incubation studies have been attributed to a high demand for  $\text{P}_i$  to support microbial growth during the incubation along with a preferential uptake of P-compounds with lighter oxygen isotopes (Zohar et al., 2010). This process will result in an accumulation of isotopically heavier phosphates in the labile P pools, which might initiate an exchange of isotopically lighter phosphates associated with oxides and minerals against those heavier phosphates from the labile P pools. This isotopic exchange can account for changes in  $\delta^{18}\text{O}_{\text{HCl P}}$  signature absent of changes in the concentration of the HCl P pool (Zohar et al., 2010). Hence, even if our approach does not allow to directly measure the re-release of P from the HCl pool into the soil solution, the HCl P pool of the



**Fig. 3.** Development of  $\delta^{18}\text{O}_{\text{P}}$  values throughout the 28 days of incubation at different soil depths. By comparison, a full equilibration of oxygen isotopes in phosphate and water during incubation would result in  $\delta^{18}\text{O}_{\text{P}}$  of ~1000‰. Asterisks indicate a significant increase of  $\delta^{18}\text{O}_{\text{P}}$  values compared to the original value at  $t_0$  ( $P < 0.05$ ). Error bars represent standard error of the mean ( $n = 3$ ).

topsoil appears to be less inert than previously suggested. We were not able to gain sufficient amounts of material for a similar incubation of biopore material. However, following the suggestion by Don et al. (2008) that biopores present 'micro-A horizons' in the subsoil, we postulate that similar assumptions as for the topsoil can be made for P in biopores, thus accounting for the changes in isotopic signature without related changes in HCl P concentrations.

For the subsoil, a change in  $\delta^{18}\text{O}_{\text{HCl P}}$  signatures within four weeks of incubation can only be observed in the upper subsoil (30–45 cm), but to a much smaller and non-significant extent. In the lower subsoil (75–105 cm),  $\delta^{18}\text{O}_{\text{P}}$  values were stable over the course of four weeks. Yet, the  $\delta^{18}\text{O}_{\text{P}}$  signature of subsoil biopores shows that even at this depth a stimulation of P cycling in the subsoil is possible in the long-term. Without additional P inputs by earthworms, however, this process apparently requires much longer time periods than realised with the incubation time in this study.

## 5. Conclusion

In the arable soil under study, biopores tended to have elevated concentrations of labile P as well as  $\delta^{18}\text{O}_{\text{HCl P}}$  values close to equilibrium even in the subsoil. The findings thus support our first two hypotheses that P cycling in biopores is increased and that this higher P cycling over long periods of time will eventually result in  $\delta^{18}\text{O}_{\text{HCl P}}$  values closer to equilibrium. We attribute this effect primarily to the re-precipitation of biologically cycled phosphate into secondary Ca-P minerals. Contrary to our third hypothesis, however, we found that Ca-P in the bulk soil outside of biopores was not completely inert, especially in the topsoil. We observed a marked increase in  $\delta^{18}\text{O}_{\text{HCl P}}$  values even within a short period of time even though HCl P concentrations did not change, indicating that in the topsoil Ca-P may participate in P turnover in the long run. It remains to be tested, though, whether long-term cycling of Ca-P can similarly account for the observed  $\delta^{18}\text{O}_{\text{P}}$  values close to equilibrium in subsoil biopores.

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